AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 1 · WORKBOOK

AI or Not?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One week. One connected evidence record.

Use the main worksheets in order. The reference bank contains alternative copies and extra practice, so you can choose support without repeating the whole week.

Keep one habit: Keep your first attempt, the evidence behind it and the correction you can explain.

MAIN LESSON · SECTION 1

Start: what did you observe?

> Try before notes; keep your later correction visible.

The speaking-clock puzzle

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- A classroom clock speaks the time. Write one thing you observed and one thing you cannot determine just from hearing it.

My starting explanation

MAIN LESSON · SECTION 12

W1 · Separate seeing from explaining

> Use the door case to separate an observation from a question about its mechanism.

Read, then explain

- A door opens when a visitor approaches. Its label says intelligent entrance. List two **observation**s and write one **mechanism** question whose answer would help classify the opening decision.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Trace the photo-library decisions

> Use the supplied facts to classify each decision and choose a useful diagnostic record.

The case facts

Given facts: a photo library uses a model fitted on labelled pictures to predict whether a new picture contains a bicycle. A written instruction puts bicycle predictions in one album. The owner can move any photo. A bicycle photo is in the wrong album; the reason has not yet been identified.

Your trace and diagnosis

- Use the photo-library case from the lesson. Classify the **prediction**, album-placement rule and owner move. A bicycle photo is in the wrong album. Name the first two records you would inspect.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Explain two possible mechanisms

> One observed turn can have more than one explanation. Name evidence that would distinguish them.

The vacuum challenge

- Maya says a vacuum must learn because it changes direction after hitting a chair. Give two different **mechanism**s consistent with that **observation**. What **evidence** would distinguish them?

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Apply the idea independently

> The greenhouse is a new case. Write one first attempt before feedback.

Read the new case, then explain

A school greenhouse estimates tomorrow's temperature using a model fitted on historical weather. A **written rule** displays CHECK PLANTS above 30 degrees C. A caretaker chooses whether to open a vent.

Identify each **mechanism** and explain why a correct CHECK PLANTS message would not prove an accurate temperature **prediction**.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ Earlier wording and repeated headings

Independent case: Investigate a new hybrid system

MAIN LESSON · SECTION 16

W5 · Design an evidence card

> Build one example that lets another person check your claim.

Create and justify

- Invent one product containing two **mechanism**s. State the supplied facts, draw an information path and write one claim the facts support and one they do not. Do not use real classmates' personal information.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

Project · Choose your sorter’s mechanism

> Make one decision and attach the evidence behind it.

Your design choice

- Choose whether the Helpful Classroom Sorter needs a person, fixed rule, or **learned model**; justify the smallest safe solution.

- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep your first attempt, the repair and a different case to revisit.

Your learning record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Check what you can explain

> Judge each item using work you can point to.

Your self-check

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Record the help you used**

REFERENCE · SECTION 2

Reference · Record the help you used

> This supplements W2; do not repeat the same completed explanation.

Use this when helpful

Use W2: Complete the photo library explanation for **Evidence** and decision **mechanism**s.
Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Are all rules outside AI?

REFERENCE · SECTION 3

Extra practice · Are all rules outside AI?

> Try the claim before discussing it.

Use this when helpful

Use the worked case for **Evidence** and decision **mechanism**s. Proposed explanation: “All rules are outside AI”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · The bus learning target

REFERENCE · SECTION 5

Reference · The bus learning target

> Use the bus explanation as a reminder, not a second independent assessment.

Use this when helpful

A bus displays a delay **prediction** learned from past journeys. A **written rule** turns the display red above 10 minutes. A driver can override the message. Identify all three **mechanism**s.

Would seeing a red display prove that learning occurred?

▼ **Extra practice · Device detective record**

REFERENCE · SECTION 6

Extra practice · Device detective record

> Use these cards and this response space if you choose the six-device investigation.

Use this when helpful

Six devices are a motion-triggered lamp, a trained photo sorter, a remote-controlled car, a calculator, an unspecified smart speaker and an unspecified robot vacuum.

Classify all six using **mechanism evidence**. Which two need **more information**?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Toy, photo app and clock

REFERENCE · SECTION 7

Extra practice · Toy, photo app and clock

> Use this as a later changed case.

Use this when helpful

A toy follows your steering buttons. A photo app learned from labelled pictures. A talking clock follows a written timetable. Which uses a **learned model**? What **evidence** supports each classification?

My answer and reason

> **Write or draw your reasoning and evidence.**

Week 1 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep predicted and observed results separate.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Revisit the bus question**

REFERENCE · SECTION 11

Reference · Revisit the bus question

> This bus question has already been used in teaching; record it as practice.

Use this when helpful

A bus displays a delay **prediction** learned from past journeys. A **written rule** turns the display red above 10 minutes. A driver can override the message. Identify all three **mechanism**s.

Would seeing a red display prove that learning occurred?

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ **Reference · Greenhouse response copy**

REFERENCE · SECTION 15

Reference · Greenhouse response copy

> This is the same W4 case. Reuse its first-attempt record.

Use this when helpful

A school greenhouse estimates tomorrow's temperature using a model fitted on historical weather. A **written rule** displays CHECK PLANTS above 30 degrees C. A caretaker chooses whether to open a vent.

Identify each **mechanism** and explain why a correct CHECK PLANTS message would not prove an accurate temperature **prediction**.

My reasoning and evidence

▼ Extra practice · Read the umbrella picture

REFERENCE · SECTION 18

Extra practice · Read the umbrella picture

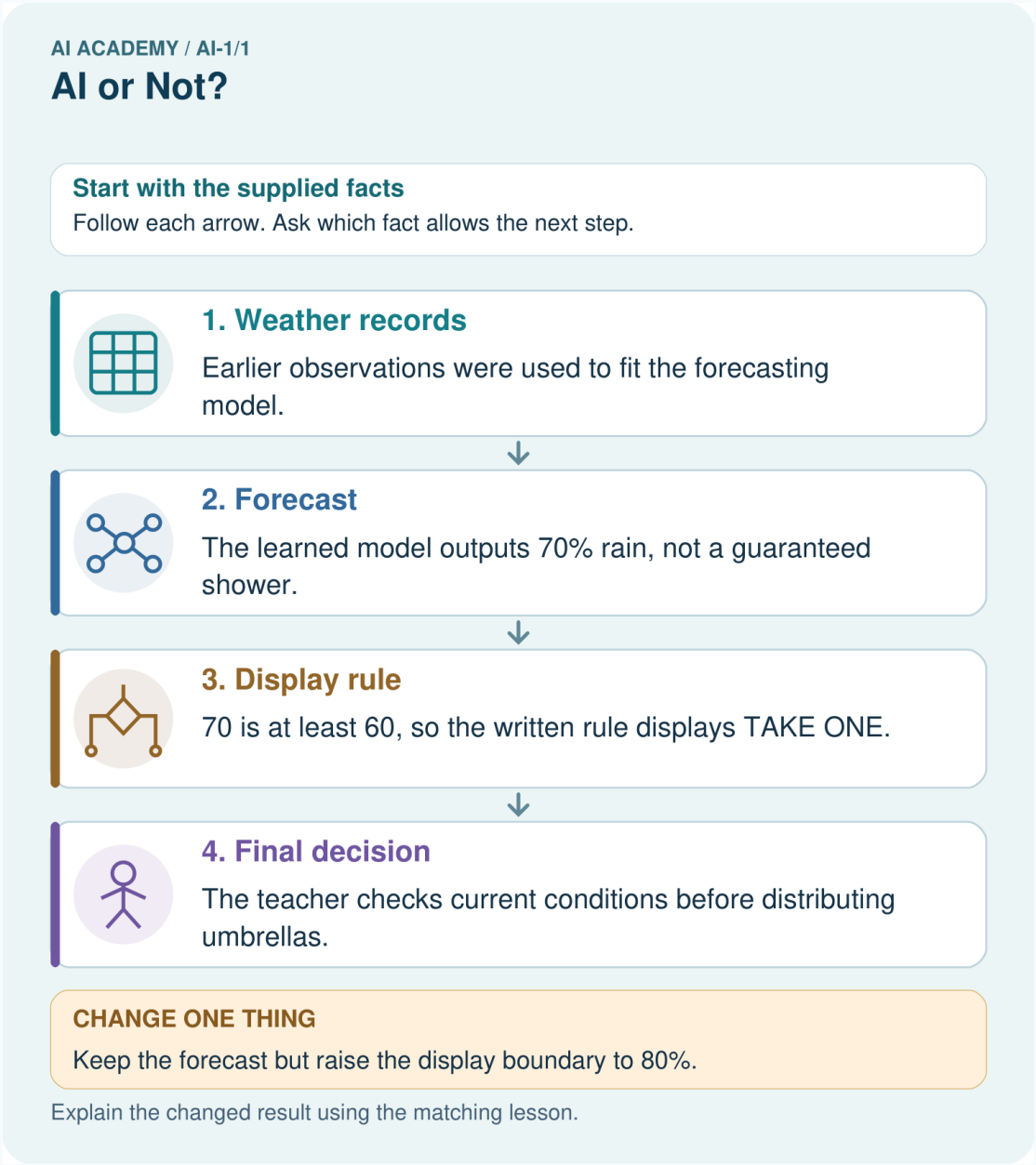
> Explain the stage and its changed boundary.

Use this when helpful

A fictional school umbrella display shows TAKE ONE before lunch. What produced that advice?

A weather model was fitted on earlier weather records; its forecast is **70%** rain.

A **written rule** shows TAKE ONE at rain forecasts of **60%** or more. The teacher decides whether to distribute umbrellas.



Follow the labelled picture with the matching explanation. Open the vector diagram

Point to the learned stage. Does the changed display prove the weather forecast changed?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Helpful Classroom Sorter, annotate which decisions follow written instructions and which, if any, were fitted from examples.

▼ **Reference · Sorter portfolio record**

REFERENCE · SECTION 19

Reference · Sorter portfolio record

> Use alongside your main project response when you need a fuller record.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

For the Helpful Classroom Sorter, annotate which decisions follow written instructions and which, if any, were fitted from examples.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Start this week's project record. Name the classroom need and explain why your proposed process fits it.

▼ Extra practice · Supported questions

REFERENCE · SECTION 20

Extra practice · Supported questions

Choose the paper or interactive version, and compare with feedback after trying.

First supported question

- A fictional school umbrella display shows TAKE ONE before lunch. What produced that advice?
- Which labelled step matches this detail? The teacher checks current conditions before distributing umbrellas.
- 1. Weather records
- 2. Final decision
- 3. Display rule
- 4. Forecast

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check after trying**

After your attempt, compare with the supported explanation: Final decision The teacher checks current conditions before distributing umbrellas. The **fitting** record supports calling the forecasting stage learned.

Comparing 70 with 60 is an explicit rule. The final action belongs to a person. Merely seeing TAKE ONE would not reveal any of these **mechanism**s; the **process** notes supply that **evidence**.

Changed boundary question

- Changed-case practice: Keep the forecast but raise the display boundary to **80%**. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. The display no longer says TAKE ONE.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after trying**

After your attempt, compare with the supported explanation: The display no longer says TAKE ONE. 70 is below 80; changing a display rule does not refit the model.

What you will keep

This is the starting lesson. Ask your teacher to model one **observation**, one **written rule** and one deciding step, then try your own example.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier prompts and notes**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

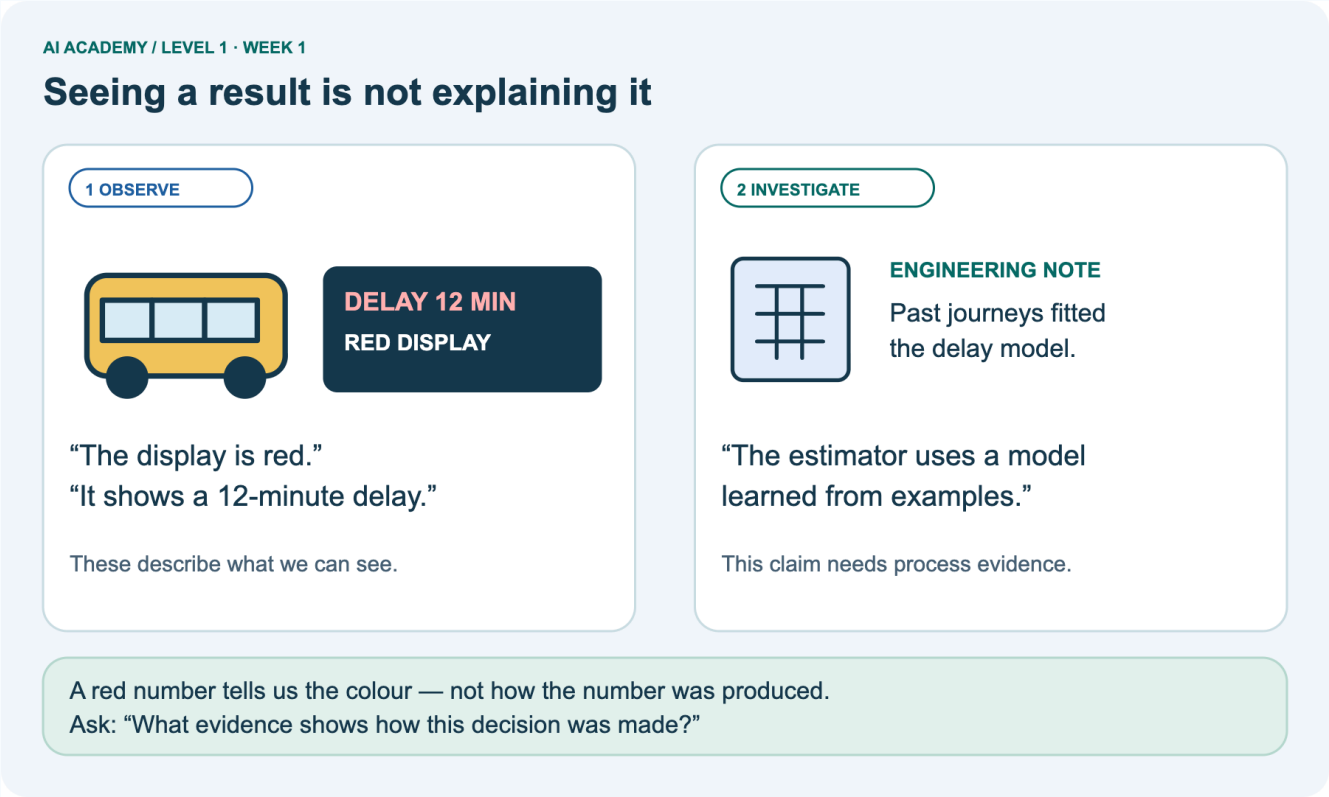
Keep your completed work

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose a relevant picture; completing all six is optional.

01. Seeing a result is not explaining it



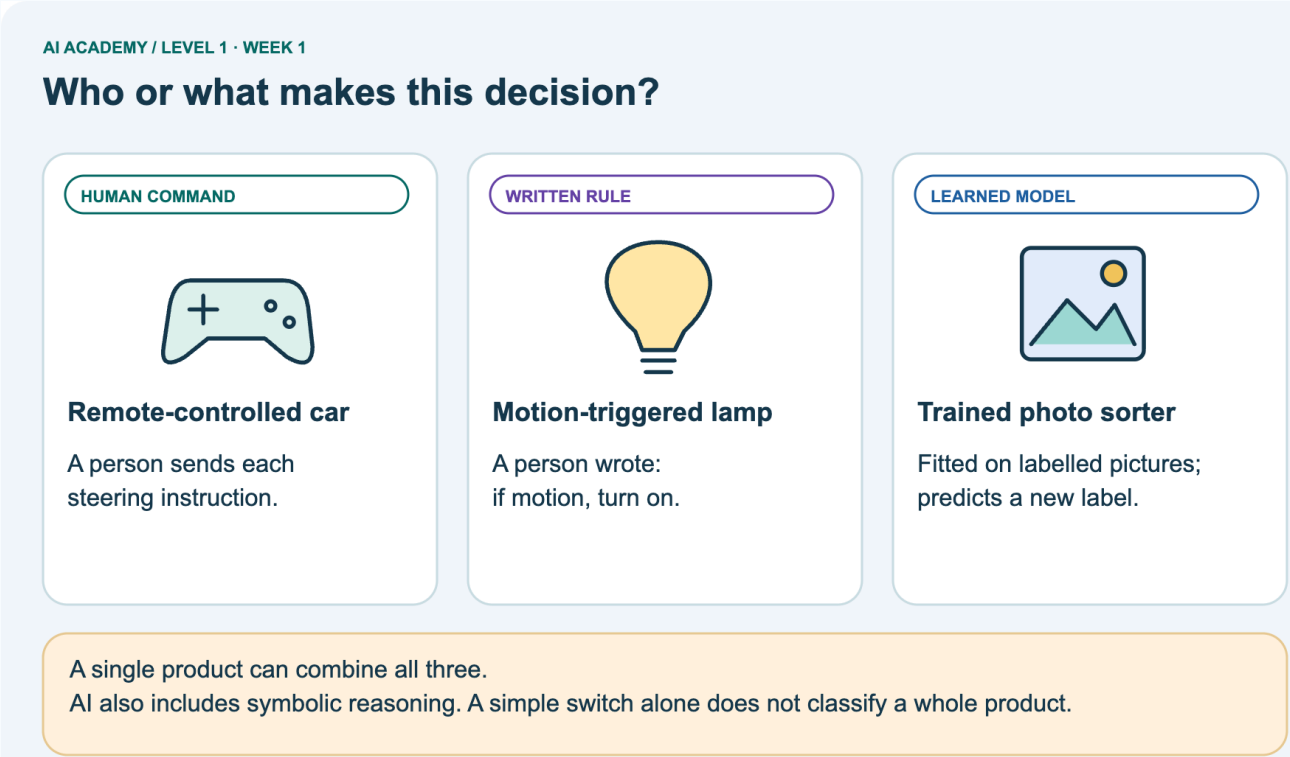
Follow the labelled picture with the matching explanation. Open the vector diagram

A bus has a red display showing a 12-minute delay. That is an **observation**. A supplied engineering note says past journeys fitted the delay model; that is **evidence** about its **mechanism**.

Name one **observation** and one claim that needs more **evidence**.

My picture explanation, deciding evidence and one limit

02. Who or what makes this decision?



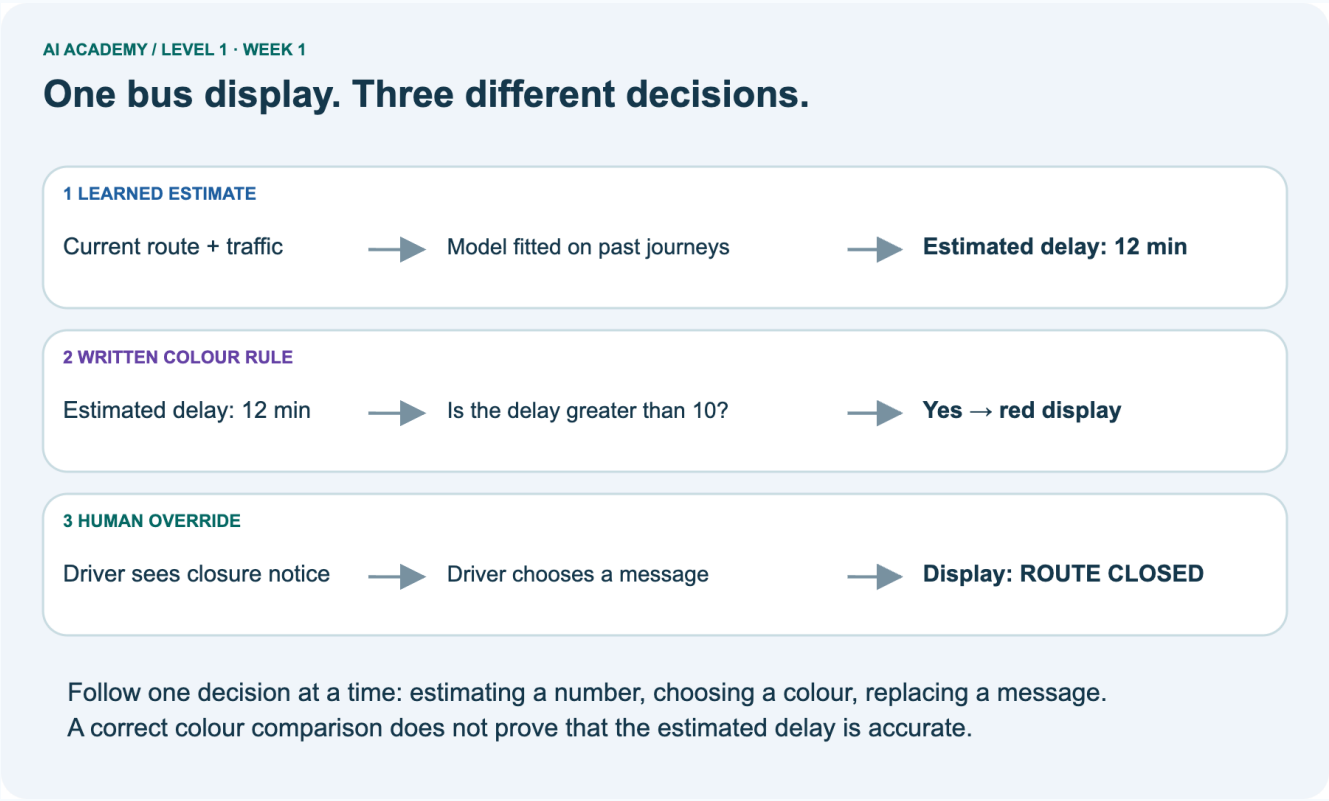
Follow the labelled picture with the matching explanation. Open the vector diagram

A remote-controlled car follows human commands, a motion lamp follows a **written rule**, and a trained photo sorter applies a **learned model**. A product can combine **mechanism**s. AI also includes symbolic reasoning.

Point to who or what makes each decision.

My picture explanation, deciding evidence and one limit

03. One bus display. Three different decisions.



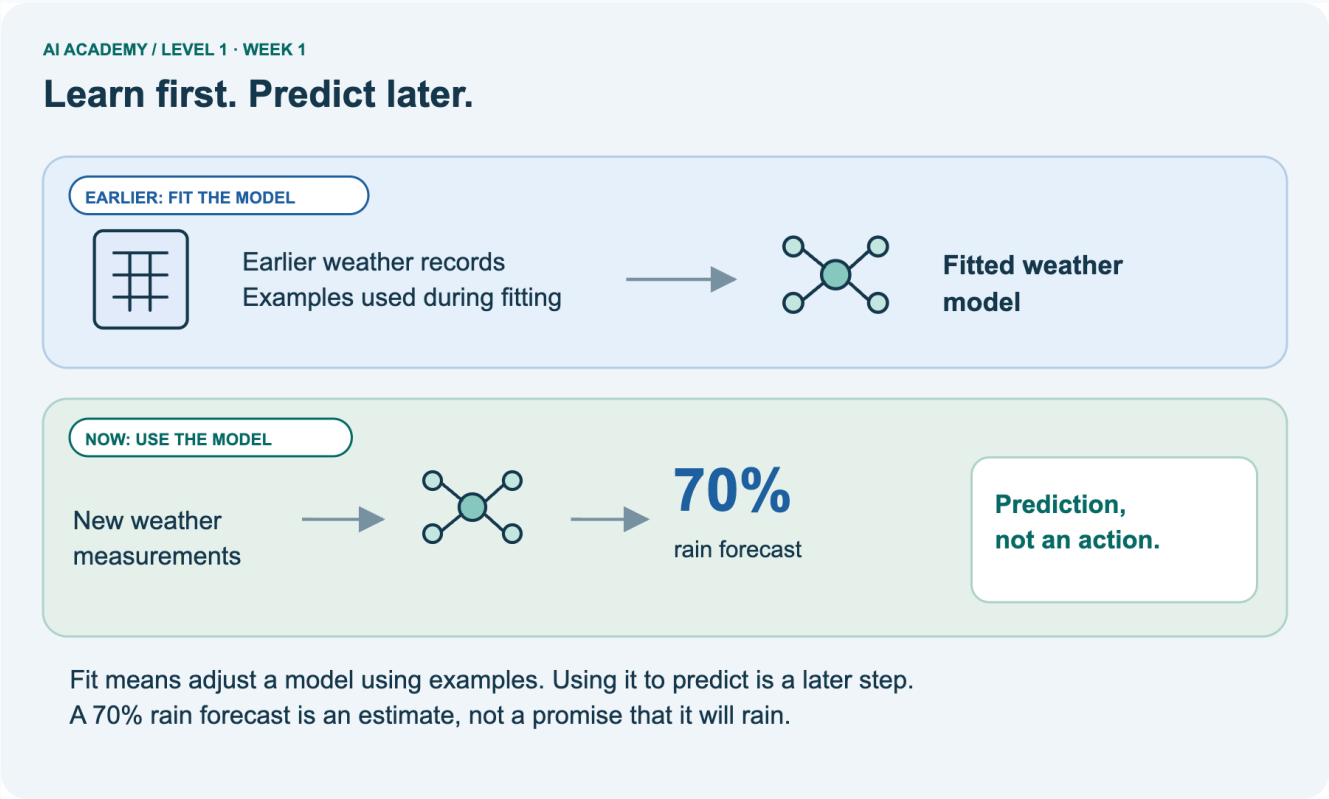
Follow the labelled picture with the matching explanation. Open the vector diagram

Three separate rows trace the bus estimate, display colour and driver override. The fitted model estimates **12 minutes**; the written greater-than-10 rule makes the display red; the driver can choose ROUTE CLOSED.

Trace the number, colour and override separately.

My picture explanation, deciding evidence and one limit

04. Learn first. Predict later.



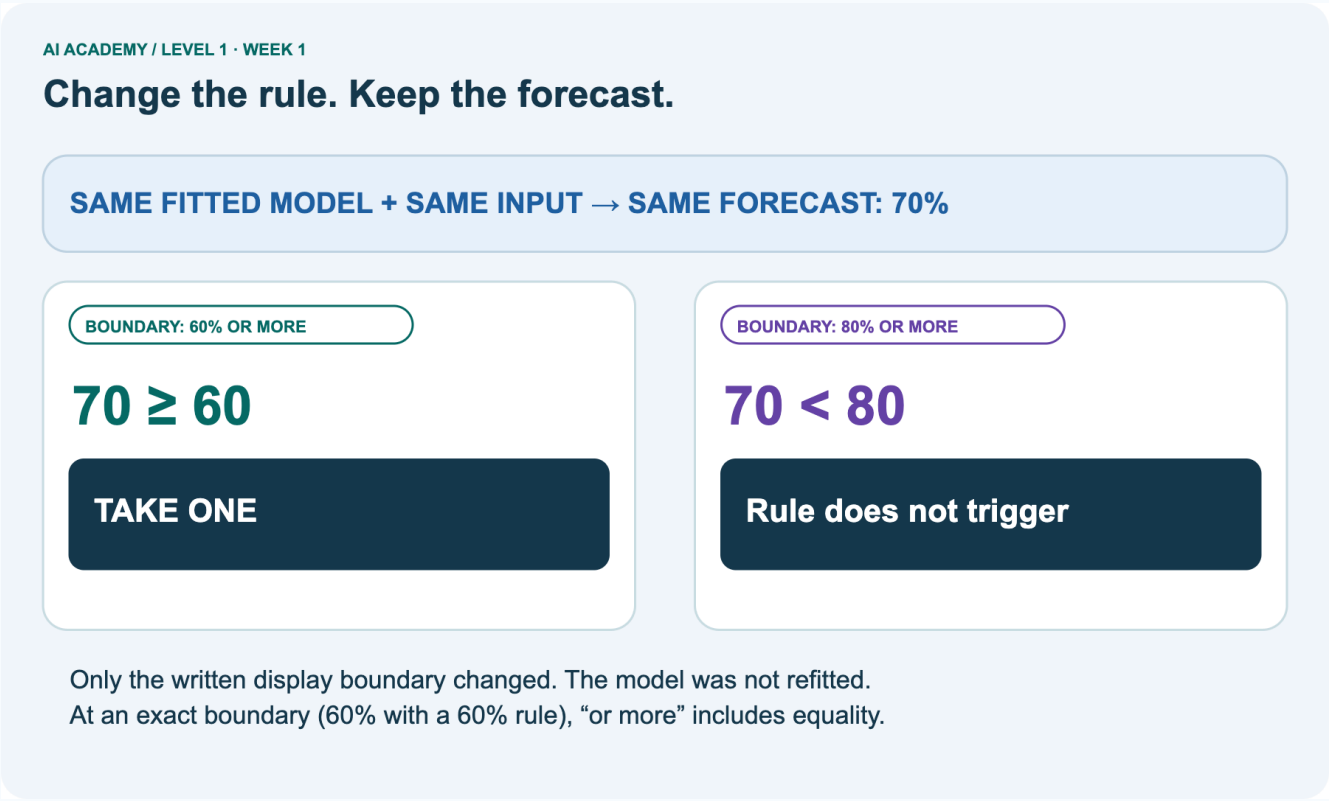
Follow the labelled picture with the matching explanation. Open the vector diagram

Earlier weather records are used to fit a model. Later, new measurements are passed to that fitted model to obtain a 70 percent rain forecast. It is an estimate, not a guarantee.

Which steps use earlier examples, and which use a fitted model?

My picture explanation, deciding evidence and one limit

05. Change the rule. Keep the forecast.



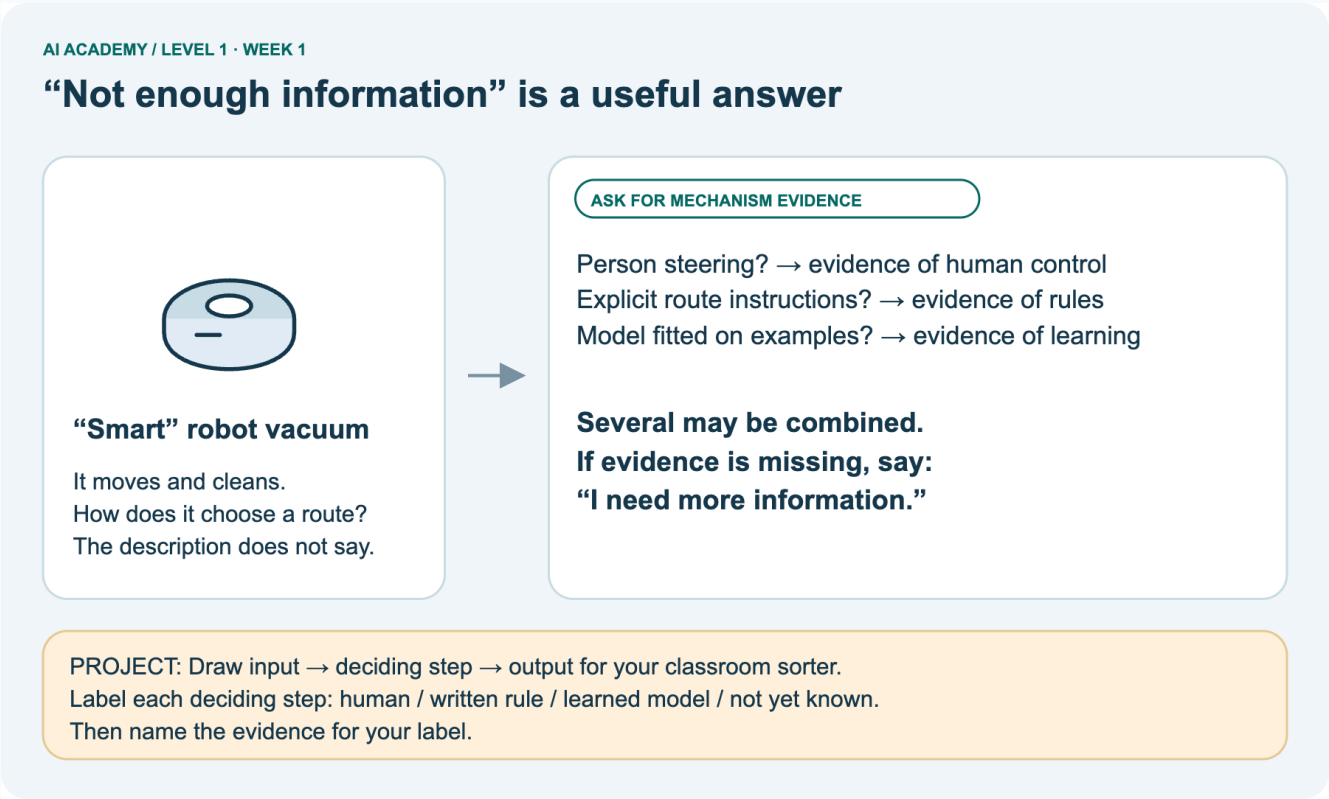
Follow the labelled picture with the matching explanation. Open the vector diagram

With a 70 percent rain forecast, a 60 percent-or-more rule displays TAKE ONE. An 80 percent-or-more rule does not trigger that message. Only the boundary changed; the fitted model and forecast did not.

Keep the forecast fixed at **70%**. What changes when the boundary becomes **80%**?

My picture explanation, deciding evidence and one limit

06. “Not enough information” is a useful answer



Follow the labelled picture with the matching explanation. Open the vector diagram

An unspecified smart robot vacuum moves and cleans, but its route **mechanism** is unknown. Ask for **evidence** of human commands, **written rule**s or model **fitting**, and allow combinations. Apply this reasoning to a classroom sorter.

Which **evidence** would change your answer about the vacuum?

My picture explanation, deciding evidence and one limit

▼ Earlier prompts and notes

01. Seeing a result is not explaining it

02. Who or what makes this decision?

03. One bus display. Three different decisions.

04. Learn first. Predict later.

05. Change the rule. Keep the forecast.

06. “Not enough information” is a useful answer

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.